Automata Project

A.9 - Present Solution - Final Project Report

Complete Project Report

Lauren Rekhelis - 743642 KHS, 9th grade

Define a problem (Design brief)

Define a problem (Design Brief)

- **Client:**
  - Who is the customer or client that is paying for the design service?
    - Automata Toy company
- **End User:**
  - Who is going to use the new product?
    - Children (girls and boys) ages of 5 to 12 years old
- **Designer(s):**
  - Who is responsible for the design?
    - Lauren Rekhelis
- **Problem Statement:**
  - What is the problem that the design team is trying to solve?
    - With knowledge of mechanisms, use the design process to design and build a mechanism or series of mechanisms which will serve as an original hand-operated toy in the US based market, producing realistic motion of a figure or objects while meeting customer required specifications.

Define a problem (Design Brief)

- **Design Statement:**
  - What does the solution have to do?
    - Design, build and test a mechanical system to automate the motion of objects (resulted in the interaction or coordinated movement of at least two separate objects) using tools and materials specified below (see Criteria and Constraints) to address a bid from the Automata Toy company that are looking for a toy solution to be used by children ages 5 to 12 years old.
- **Criteria and Constraints:**
  - What criteria does the solution have to meet?
    - The design of the system must include the wood box.
    - The mechanical system must be human powered.
    - The cost of the system must not exceed $15.00 (excluding cost of box).
    - The system must produce repetitive motion of figures or objects related to a theme.
    - The assembled hand-operated toy must fit completely within a rectangular prism that measures 6 in. x 8 in. x 12 in. when in a static state.

Define a problem (Design Brief)

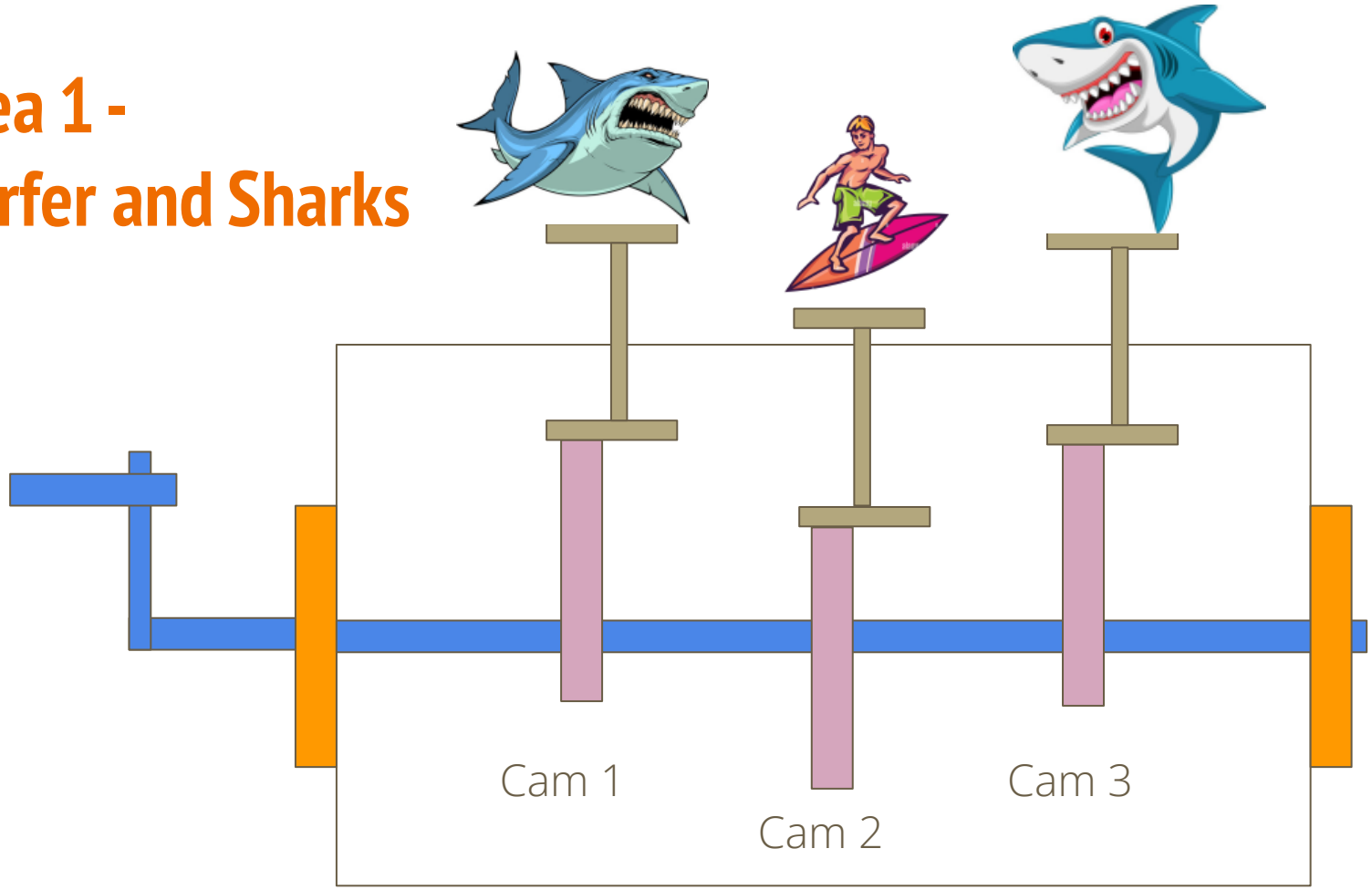
- **Criteria and Constraints (cont.):**
  - What criteria does the solution have to meet?
    - The axle must pass through the wood box. Turning the axle must convert the rotational motion of the axle to linear motion of objects outside of the wood box.
    - The mechanical system must include at least two cams.
    - The system must include components manufactured from at least four different materials, including the wood used to construct the box. Connectors or adhesives are not considered to be materials.
    - The automata objects should display realistic motion of the real objects they represent.
    - The product must be safe.

Define a problem (Design Brief)

- **Criteria and Constraints (cont.):**
  - What criteria does the solution have to meet?
    - The toy must meet quality and reliability specifications:
      - When a force of 4.8 Newtons is exerted horizontally on any component and in any direction at a height of 5 cm above the surface, the system must remain solidly founded (no part of the design lifts off of the surface).
      - The system must operate as designed for at least one minute at a sustained speed of one revolution per second without noticeable damage to any component of the design.
      - There must be no visible “slip” or “disconnect” between components in all cycles.
    - The design must allow for the reduction of friction during operation:
      - Between the fingers and the handle
      - Between the axle and its support
      - Between the follower rod and the components that guide the rod

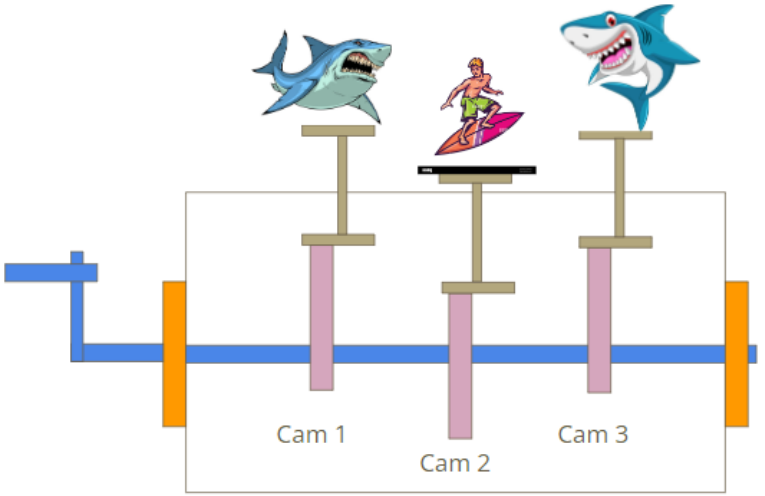
Generate concepts

Idea 1 -
Surfer and Sharks

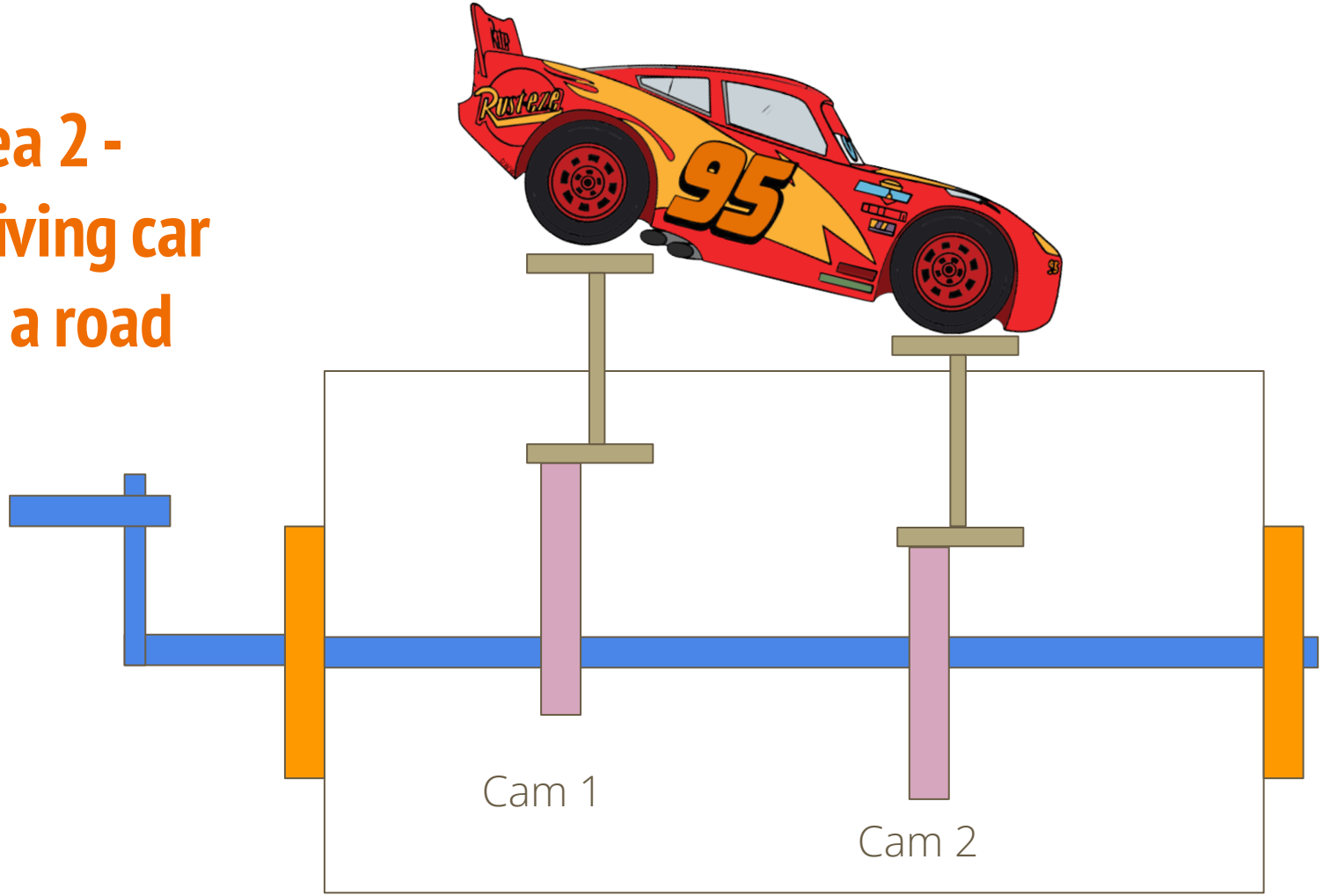


Idea 1 - Surfer and Sharks

- First idea for automata project consist of a surfing person and sharks.
- The toy will include three (3) cams with two (2) sharks and one (1) person surfing.
- The sharks and the surfer will move up and down so it looks like the person is surfing and the sharks are swimming.
- Relevant background will be added to create realistic scenario.

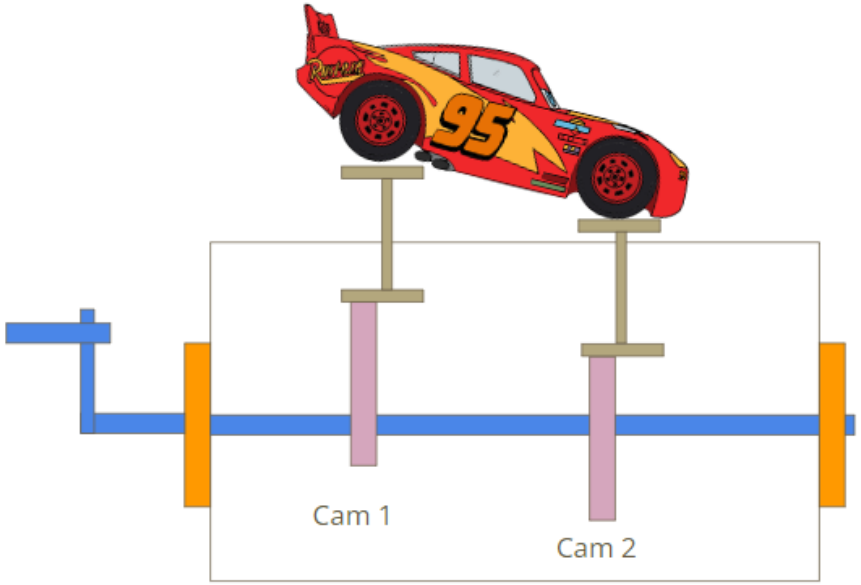


Idea 2 - Driving car on a road

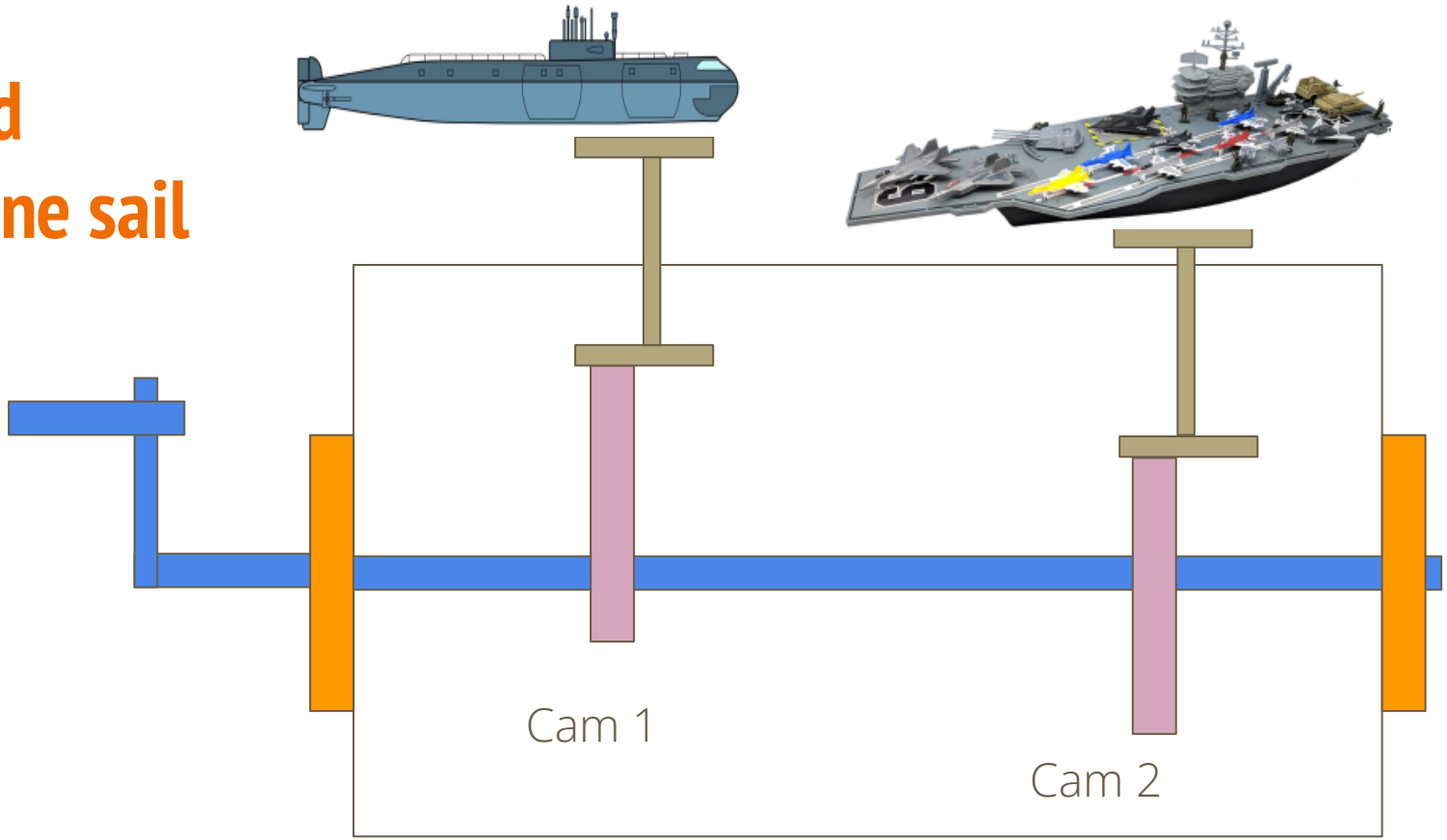


Idea 2 - Driving car on a road

- Second idea for automata project consist of a driving car driving on a mountain or unpaved road.
- The toy will include two cams connected to the car, one in the front and one in the back of the car.
- The cams will be used to make the car look likes its driving on unpaved road.

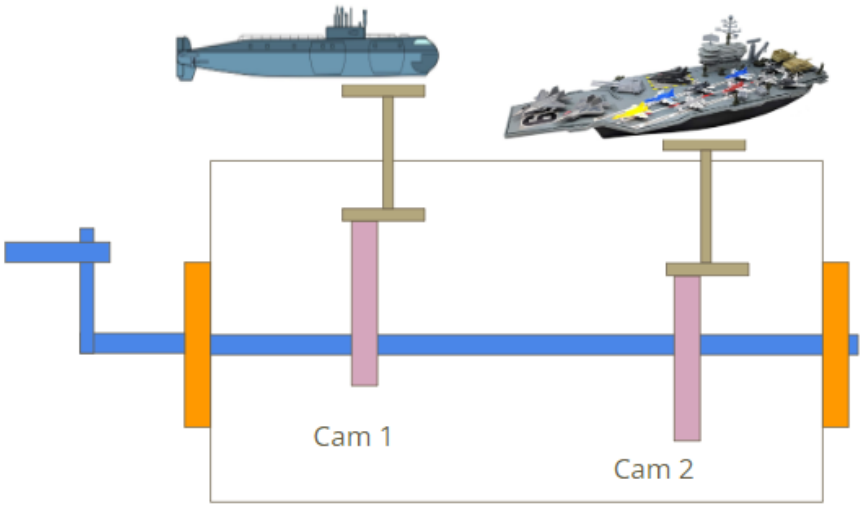


Idea 3 - Boat and submarine sail



Idea 3 - Boat and submarine sail

- Third idea for automata project consist of a aircraft carrier and submarine sailing in the sea.
- The toy will include two cams that will make the two different type of boat look like sail in the open sea, while one cam will be connected to aircraft carrier and other cam will be connected to submarine.
- The toy will include relevant background picture to add more realistic scenario.



Decision matrix

- Decision matrix was done in separate Google Sheet and attached here for review.
- Based on the the results, the Idea 1 - Surfer and Sharks - was selected for implementation.
- The scoring was done by compare of each idea in specific criteria by giving score 1, 2 or 3 based on key below: 3 - best, 1 - worst.

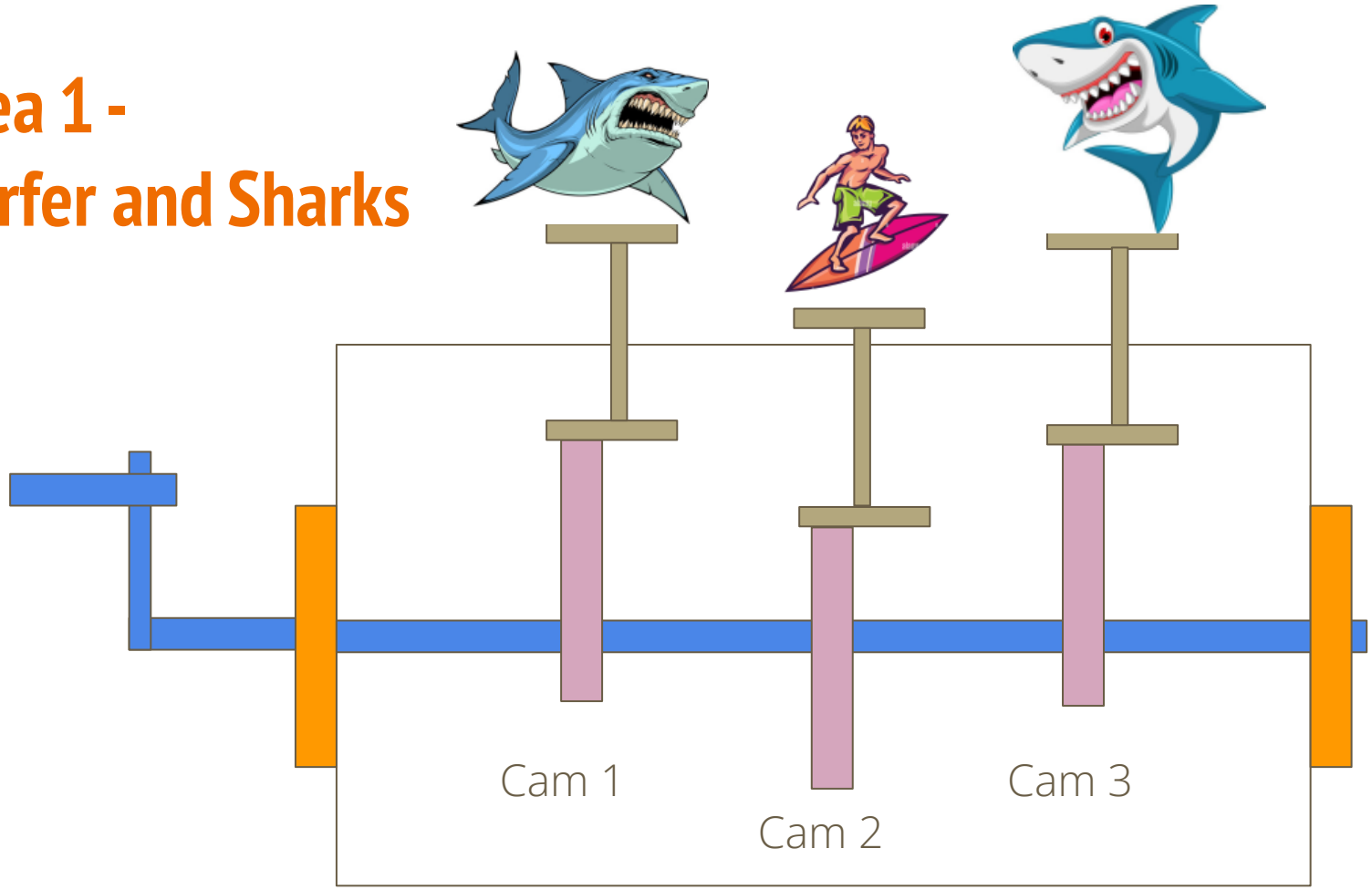
Decision Matrix - Automata project

Automata project - Criteria - Decision matrix

IDEAS	Complexity	Meet interest for both boys and girls	Cost	Safe	Good for kids ages 5-12	TBD 3			TOTALS
Surfer and Sharks	1	3	2	2	3				11
Driving car on road	2	2	1	3	2				10
Boat and submarine sail	3	1	3	1	1				9
		3 BEST	2	2	1 WORST		2 YES	1 NO	

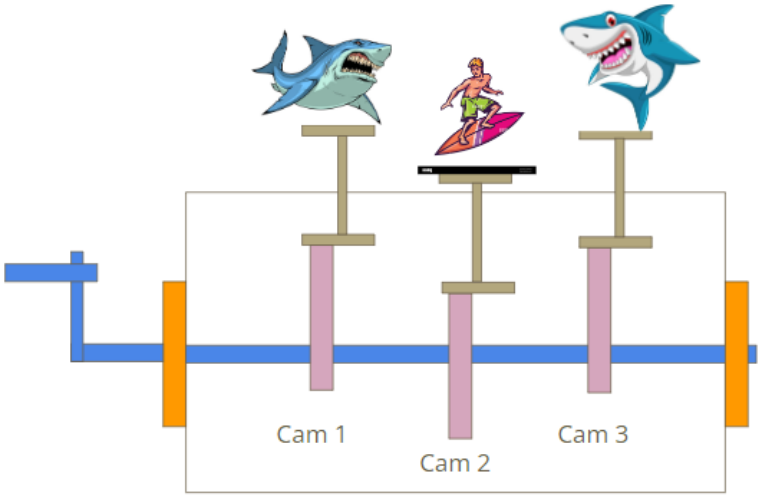
Develop a solution - Build a Box

Idea 1 -
Surfer and Sharks

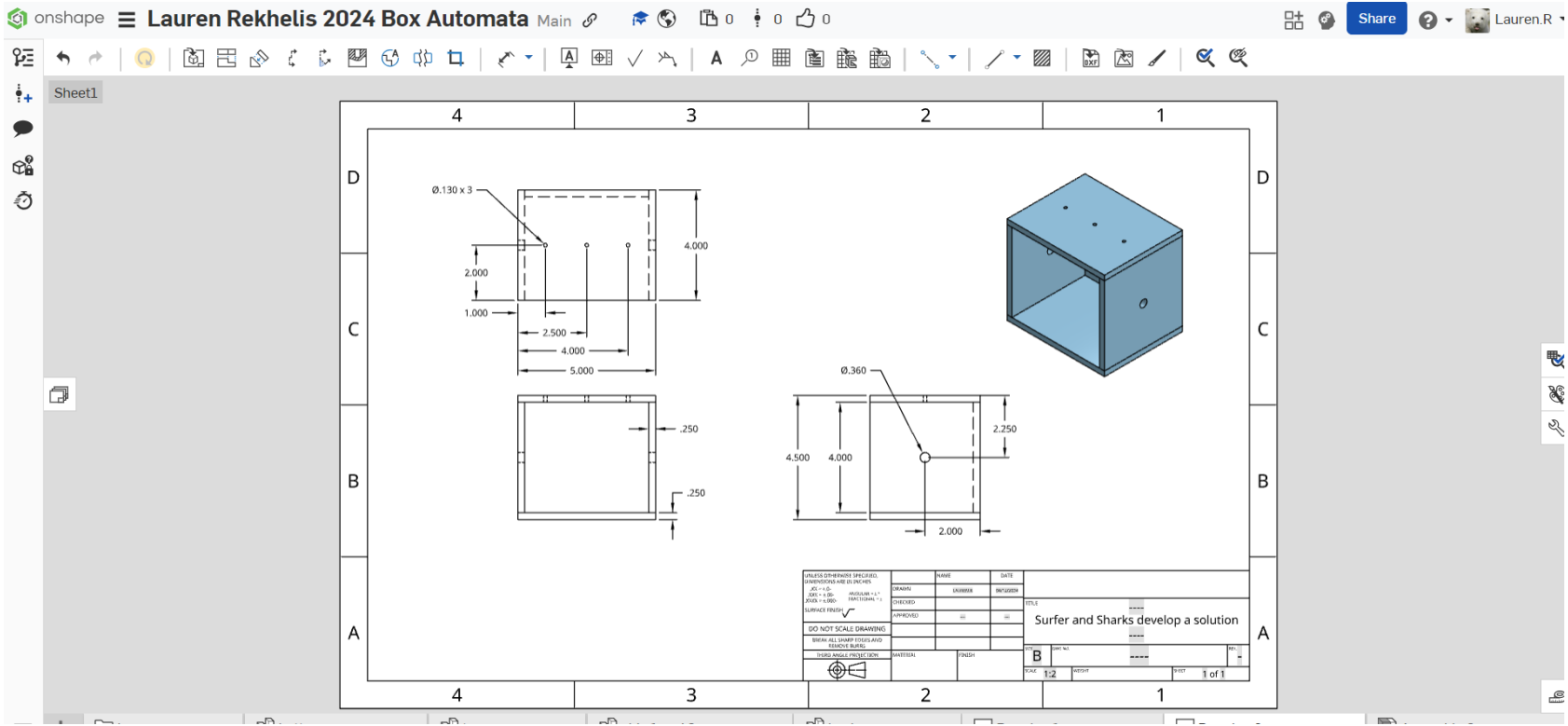


Idea 1 - Surfer and Sharks

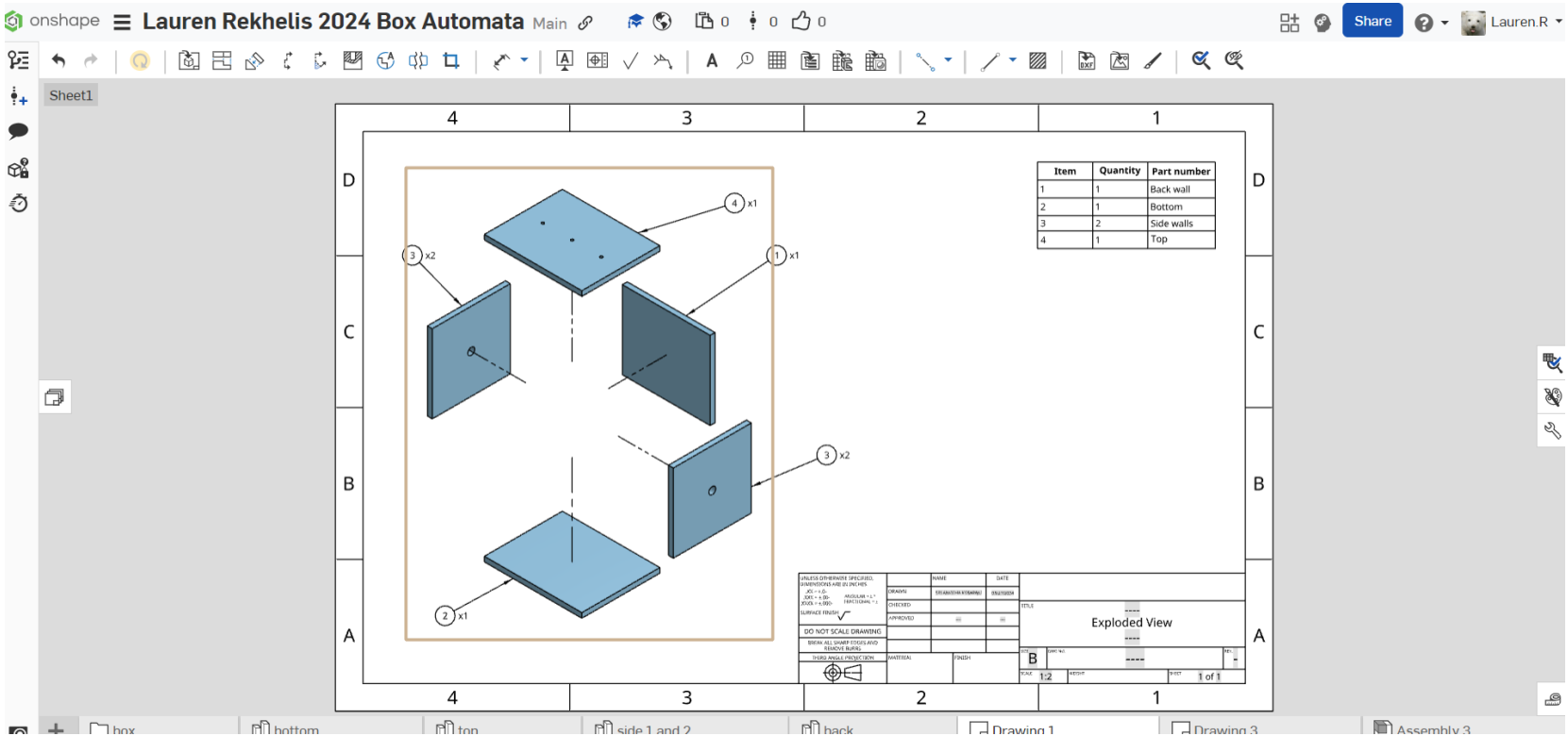
- First idea for automata project consist of a surfing person and sharks.
- The toy will include three (3) cams with two (2) sharks and one (1) person surfing.
- The sharks and the surfer will move up and down so it looks like the person is surfing and the sharks are swimming.
- Relevant background will be added to create realistic scenario.



Fully dimensional drawing with shaded isometric view

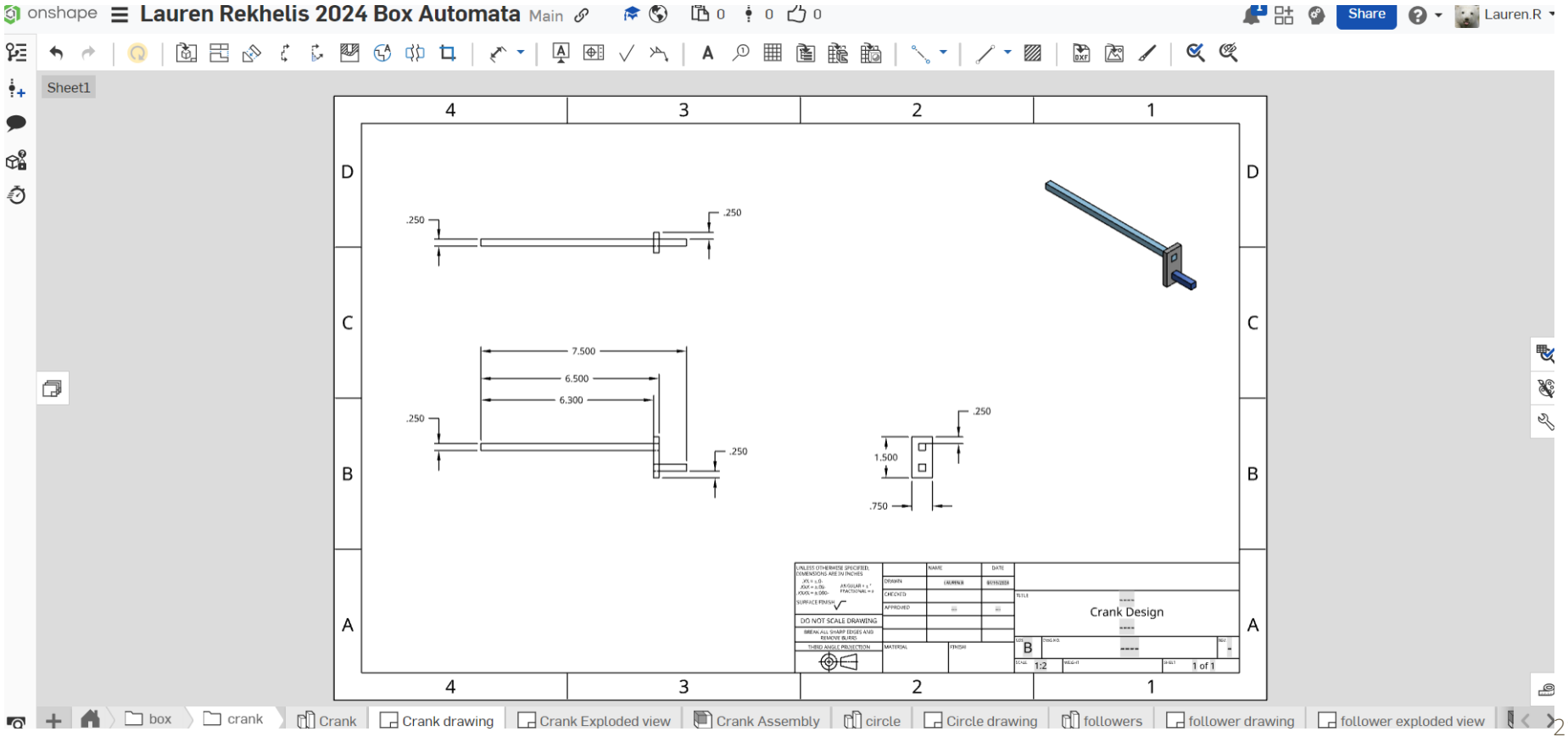


Exploded view of the assembly with parts BOM labeled

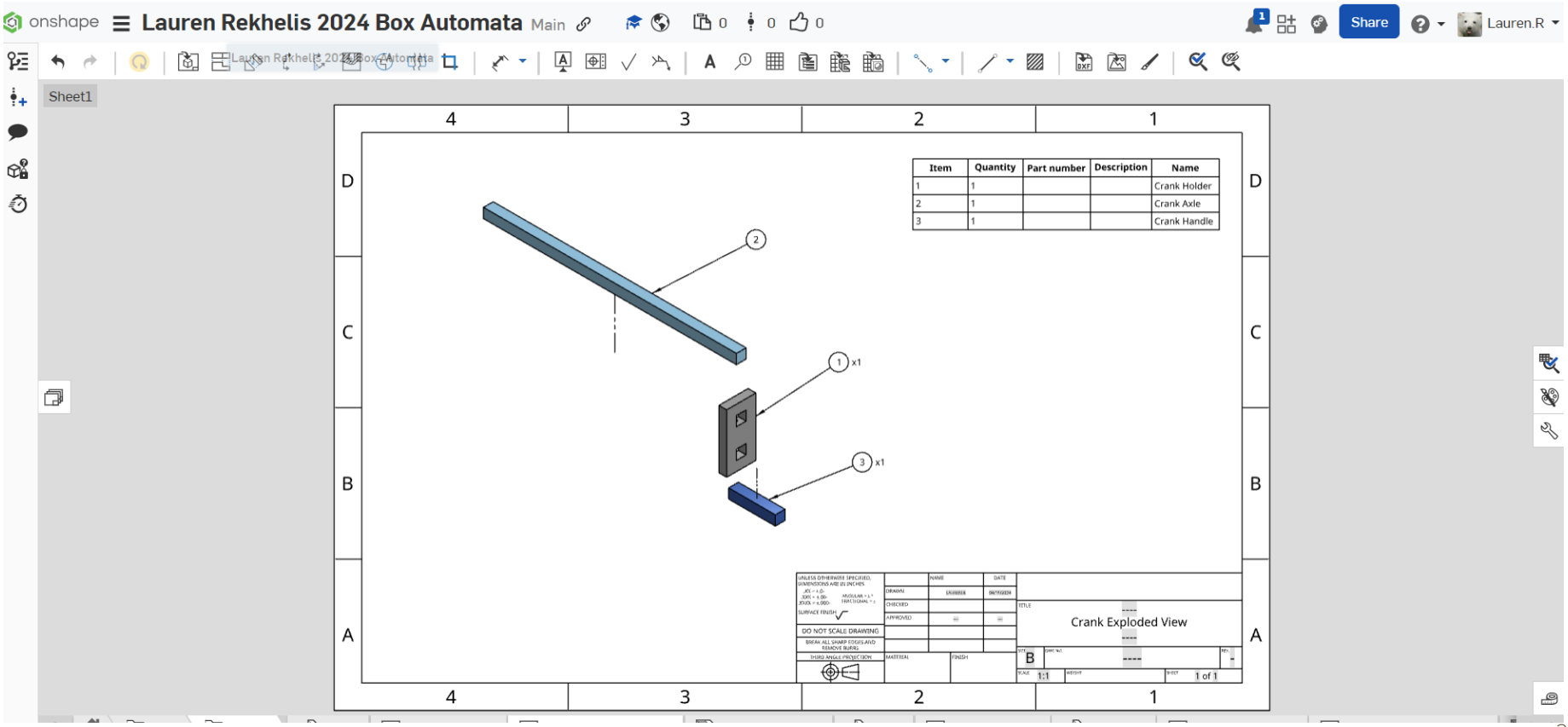


Develop a solution - Follower and Crank

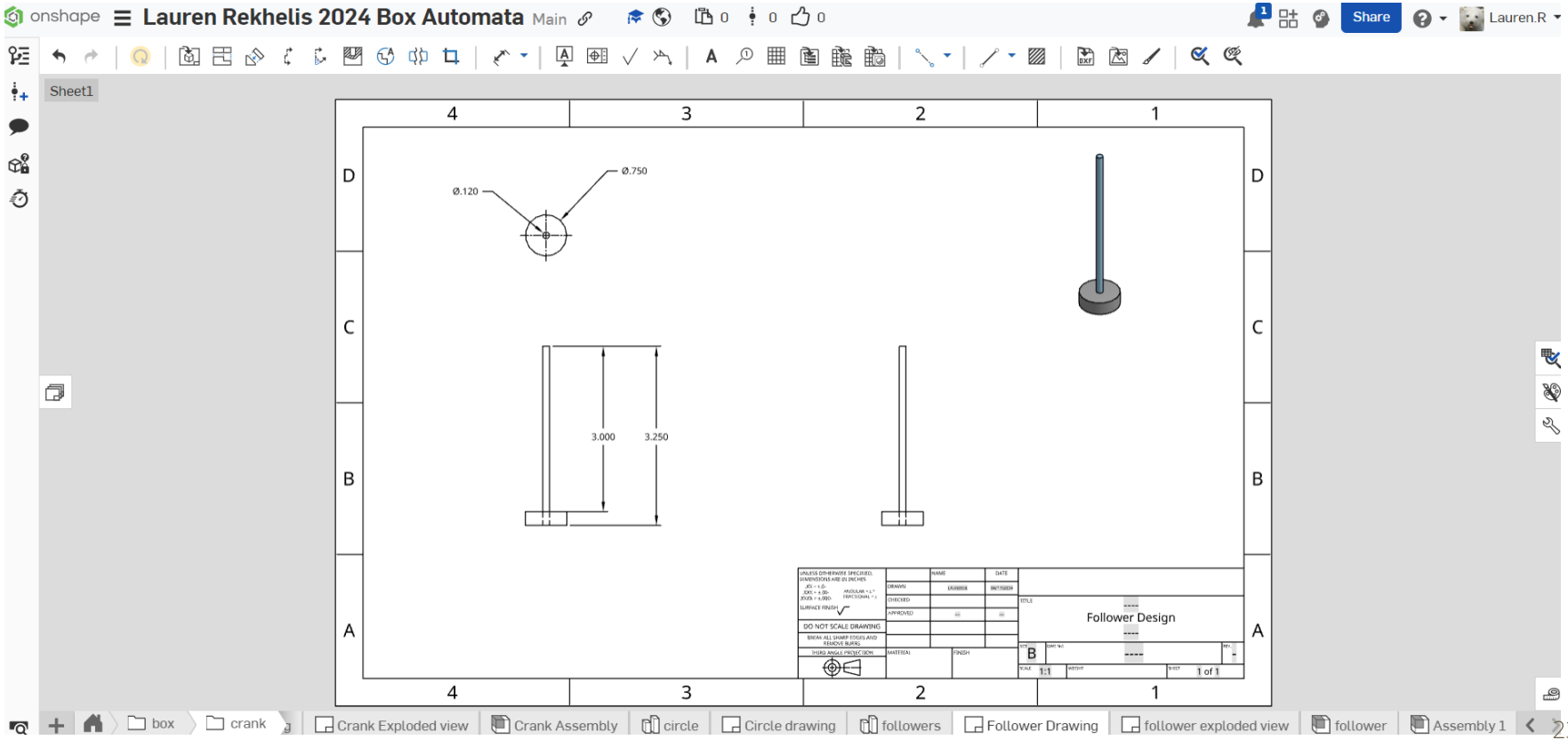
Crank - Fully dimensional drawing with shaded isometric view



Crank - Exploded view of the assembly with parts BOM labeled



Follower - Fully dimensional drawing with shaded isometric view

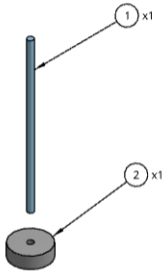


Follower - Exploded view of the assembly with parts BOM labeled

onshape ☰ Lauren Rekhelis 2024 Box Automata Main 0 0 0 Share Lauren.R

Sheet 1

Item	Quantity	Part number	Description	Name
1	1			Wood Stick
2	1			Follower Cap



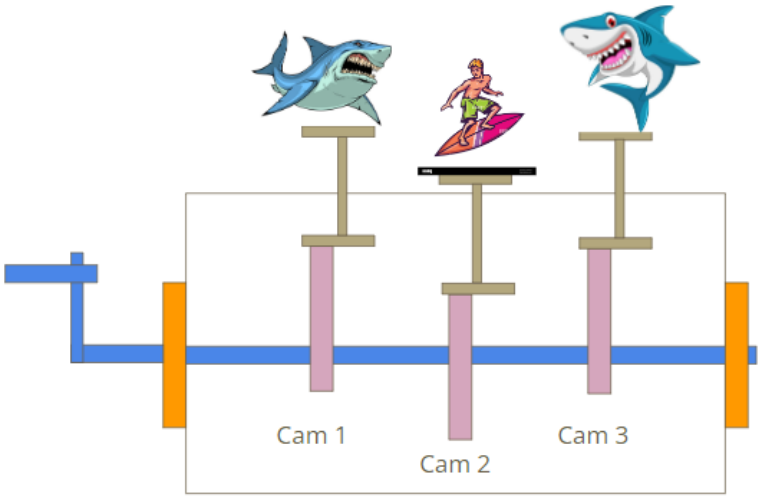
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
DRAWN LAUREKH 06/12/2024
CHECKED
APPROVED
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
MATERIAL
FINISH
Follower Exploded View
B
1:1
1 of 1

Develop a solution - Design a Cam

To create a desired vertical motion

Design your own cam - Surfer and Sharks

- The automata project toy consist of a surfing person and sharks.
- The toy will include three (3) cams with two (2) sharks and one (1) person surfing.
- The sharks and the surfer will move up and move down to create imitation of the surfing person and the sharks who are swimming and trying to chase the surfer.
- Relevant background will be added to create realistic scenario.
- Each one of three (3) cams will move asynchronous to each other to make the toy more entertaining.
- There will be a use of ECCENTRIC cam with offset to achieve desired movement in the design.

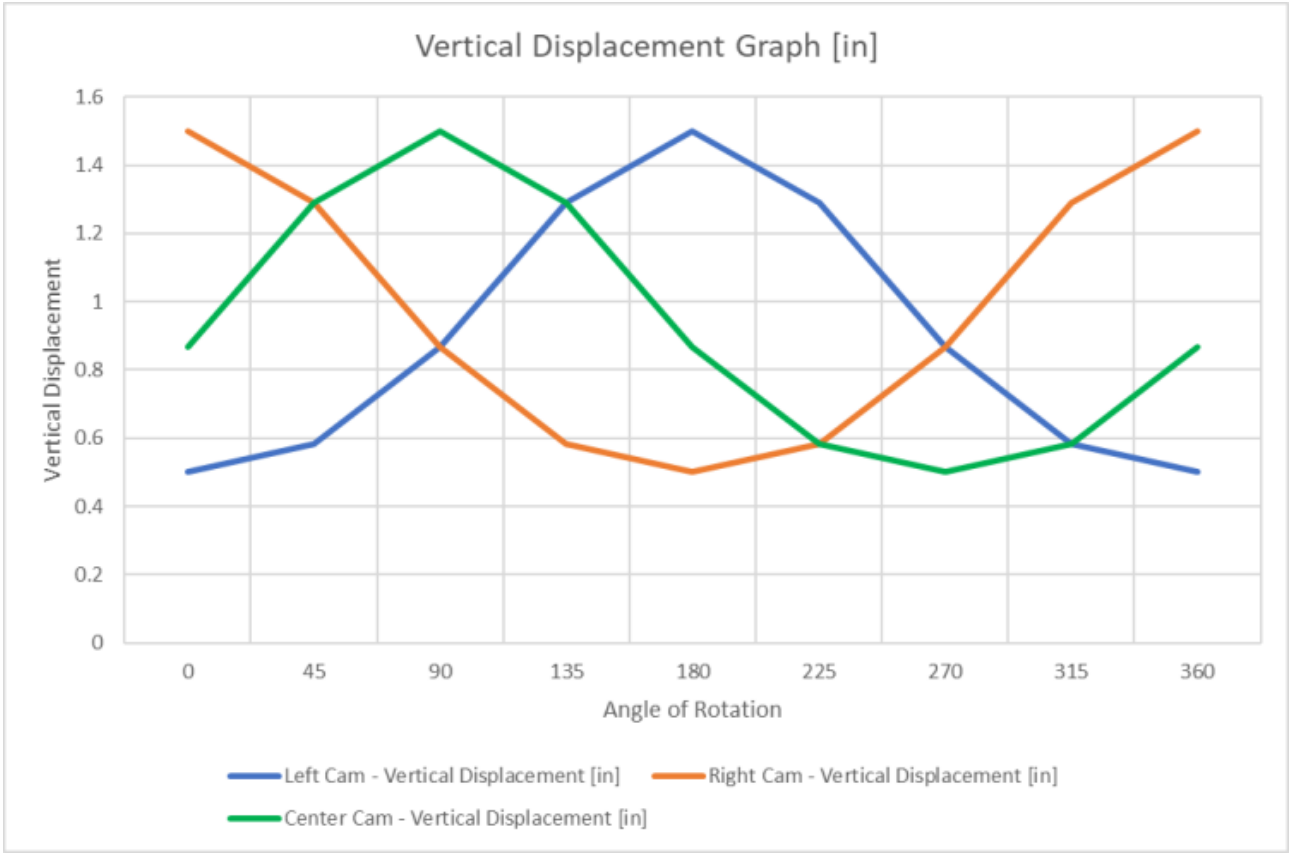


Design your own cam - Vertical Displacement (data)

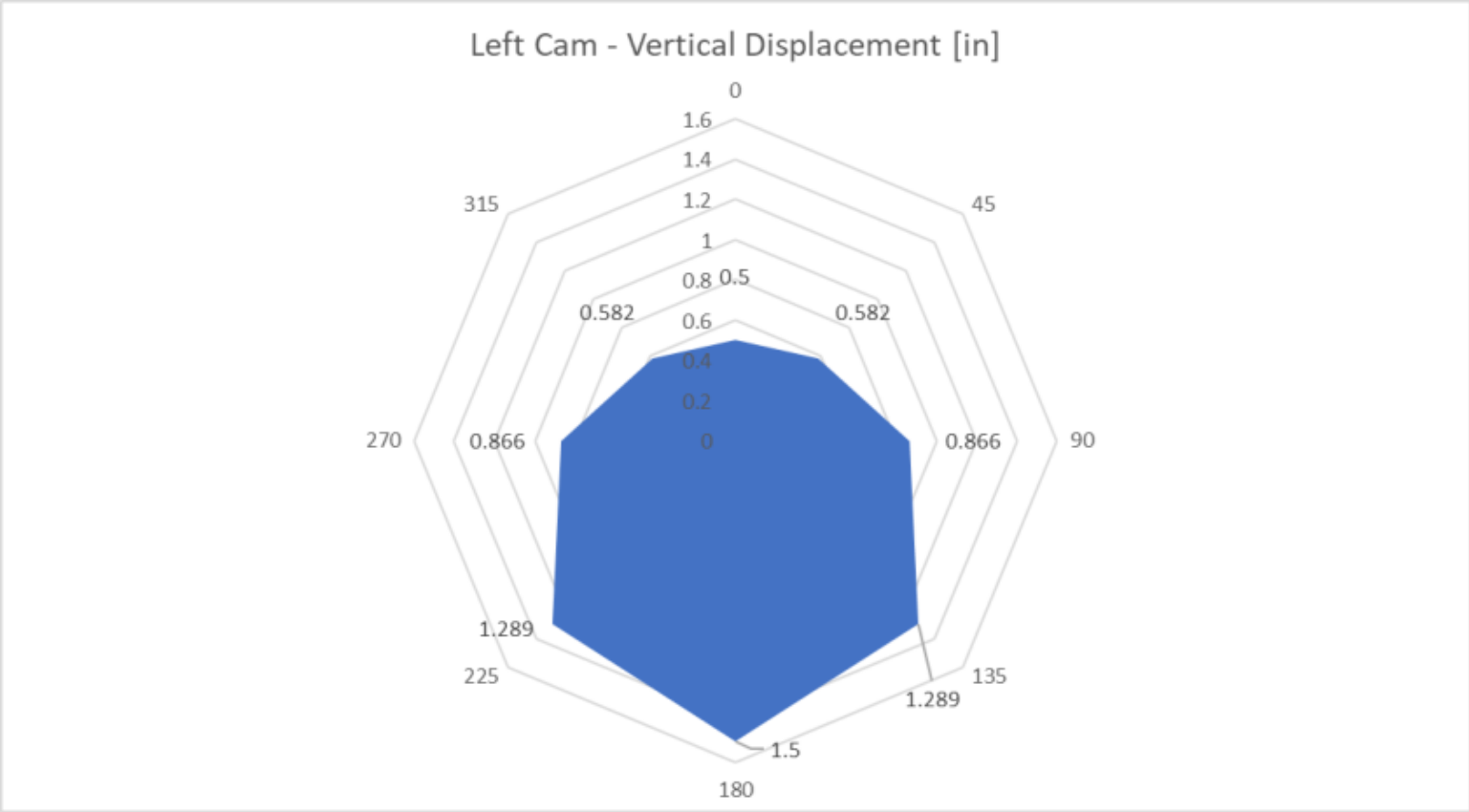
- Three cams are required to operate 90 deg to each other that will lead to vertical motion. Each cam will be 2 in diameter.

Vertical Displacement Data			
Angle [deg]	Left Cam - Vertical Displacement [in]	Center Cam - Vertical Displacement [in]	Right Cam - Vertical Displacement [in]
0	0.5	0.866	1.5
45	0.582	1.289	1.289
90	0.866	1.5	0.866
135	1.289	1.289	0.582
180	1.5	0.866	0.5
225	1.289	0.582	0.582
270	0.866	0.5	0.866
315	0.582	0.582	1.289
360	0.5	0.866	1.5

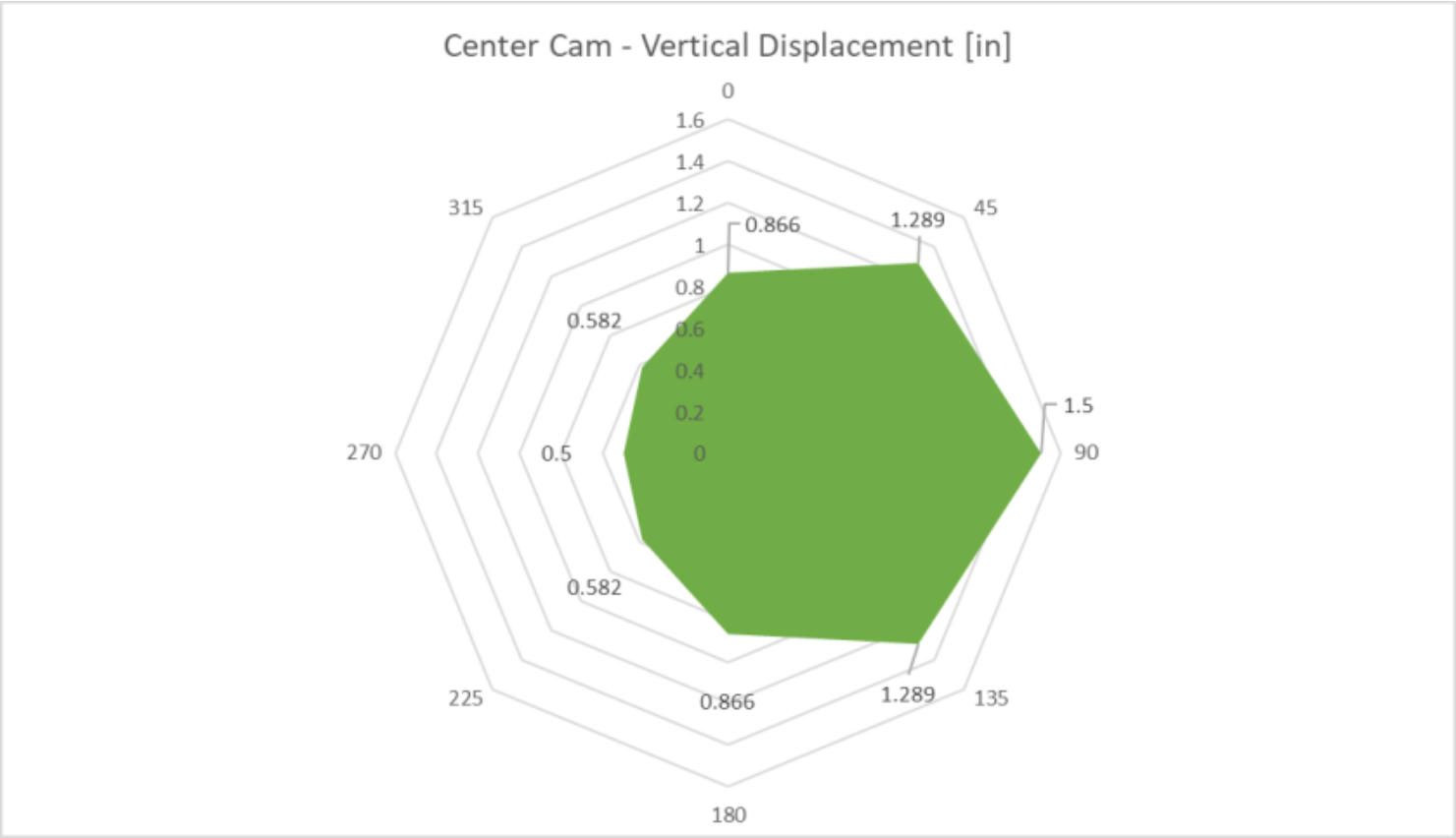
Design your own cam - Vertical Displacement (graph)



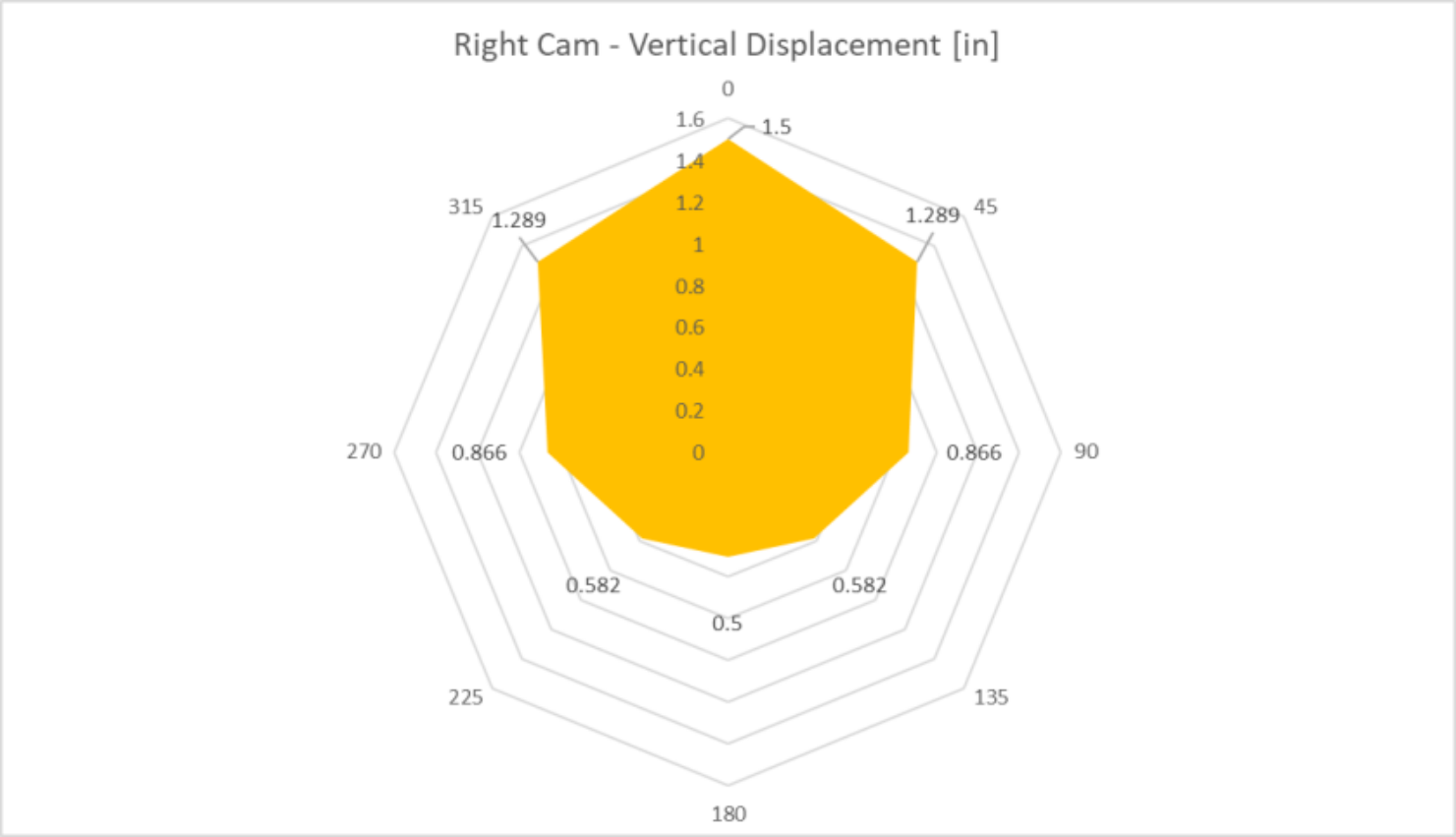
Cam profile on Polar Grid - Left Cam



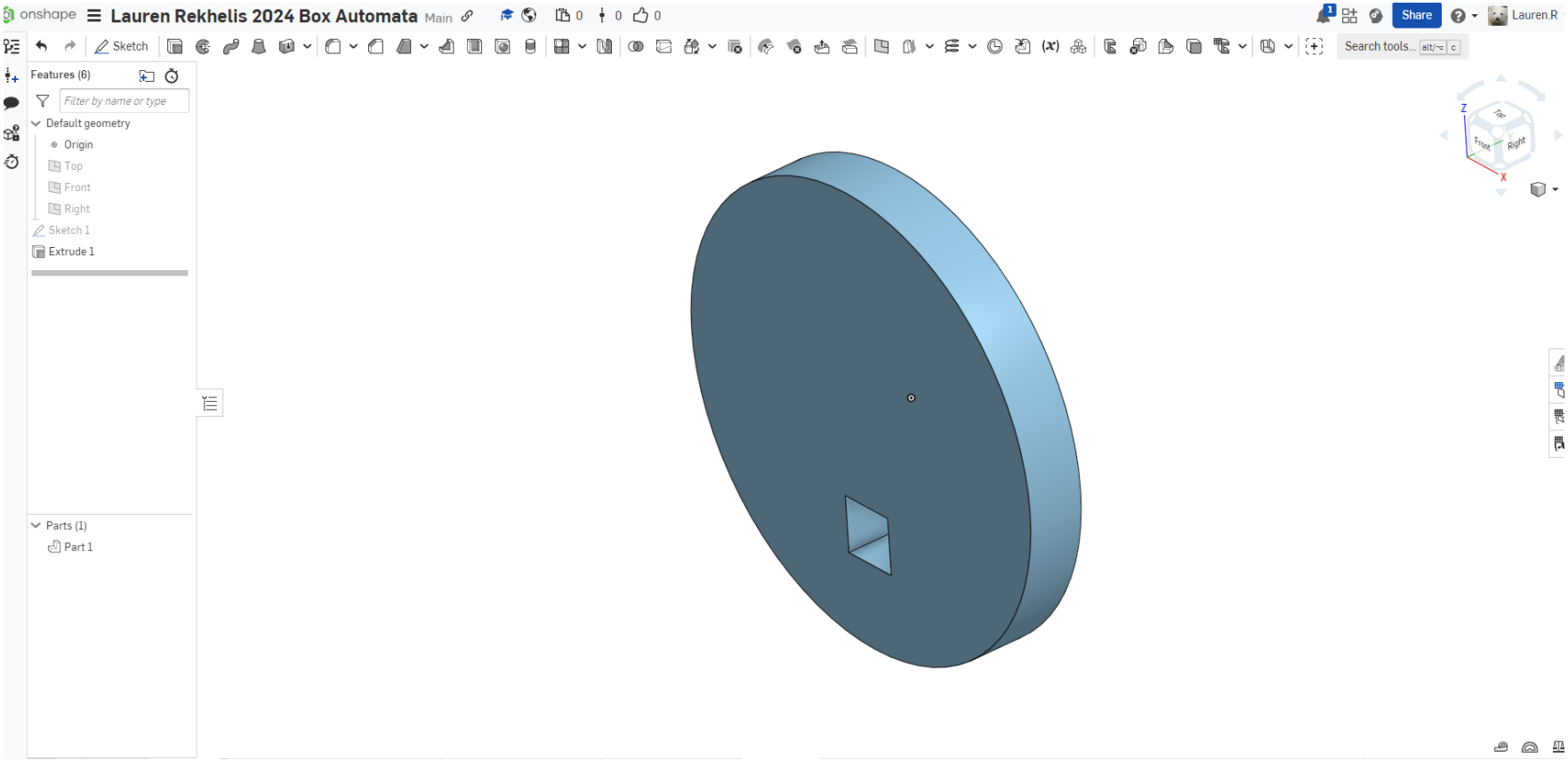
Cam profile on Polar Grid - Center Cam



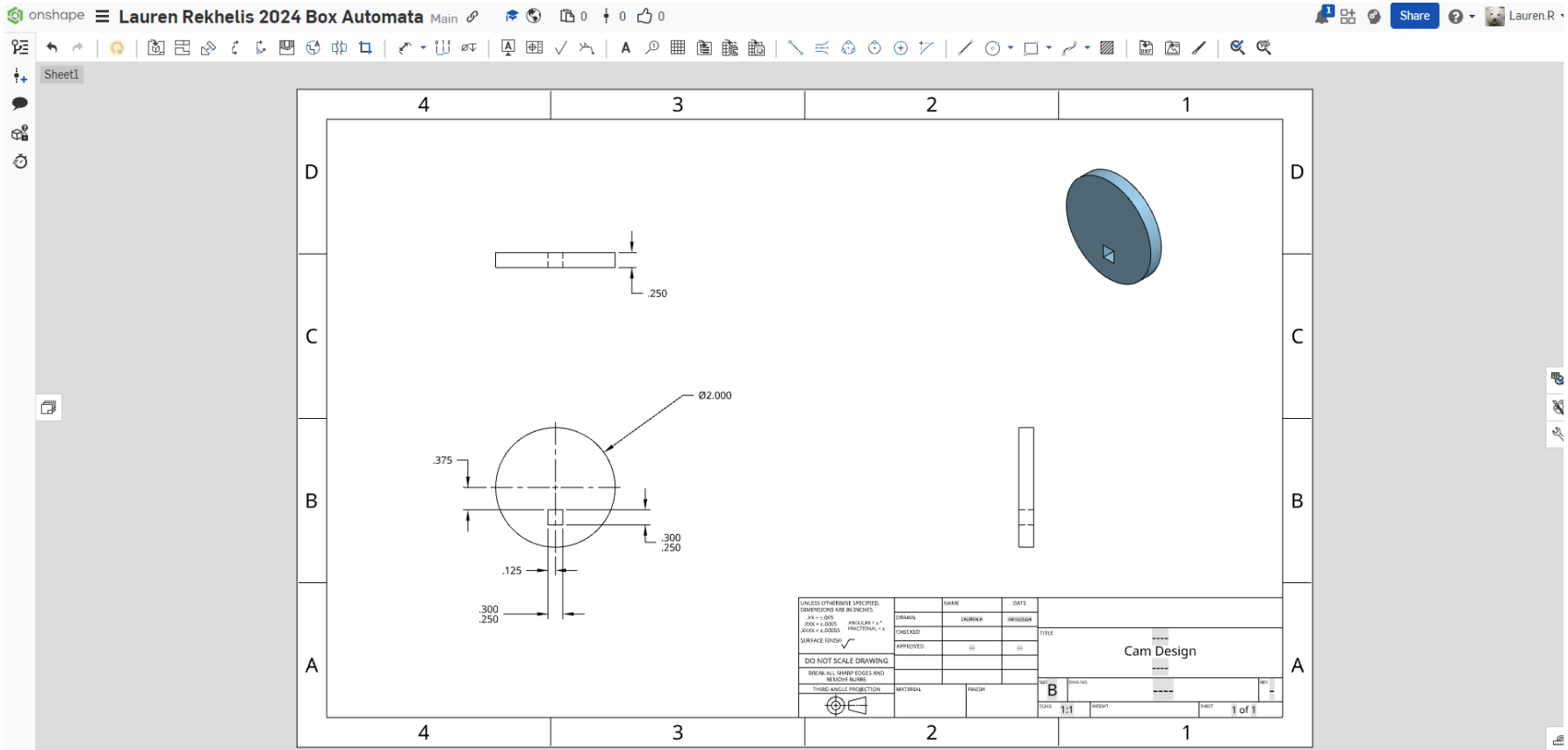
Cam profile on Polar Grid - Right Cam



Cam Design - 3D model



Fully dimensional multiview drawing, shaded isometric view



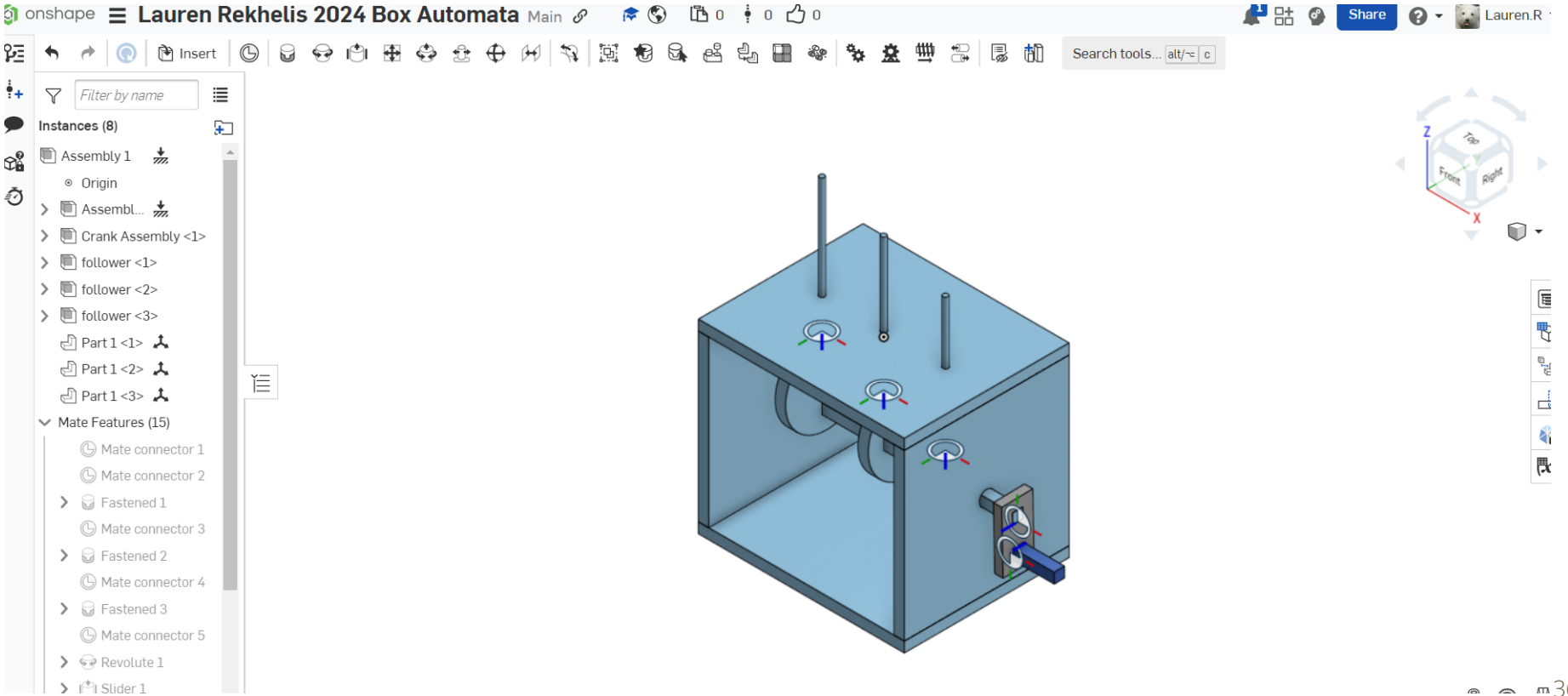
Describe why the motion of a follower acted on by a cam is periodic motion

- The periodic motion of a follower acted on by a cam is a result of the combination of a repeating cam profile and the constant rotation of the camshaft.
- The period of the motion is determined by the shape of the cam and the speed at which it rotates.
- The rotational movement of the camshaft ensures that the follower experiences a cyclical motion.
- The follower's displacement follows the contours of the cam, resulting in a motion that repeats itself with each revolution of the camshaft.
- Since the camshaft rotates continuously at a constant speed and the cam's shape remains consistent, the motion of the follower repeats itself with each rotation of the camshaft. This repetitive nature of the motion defines its periodicity.

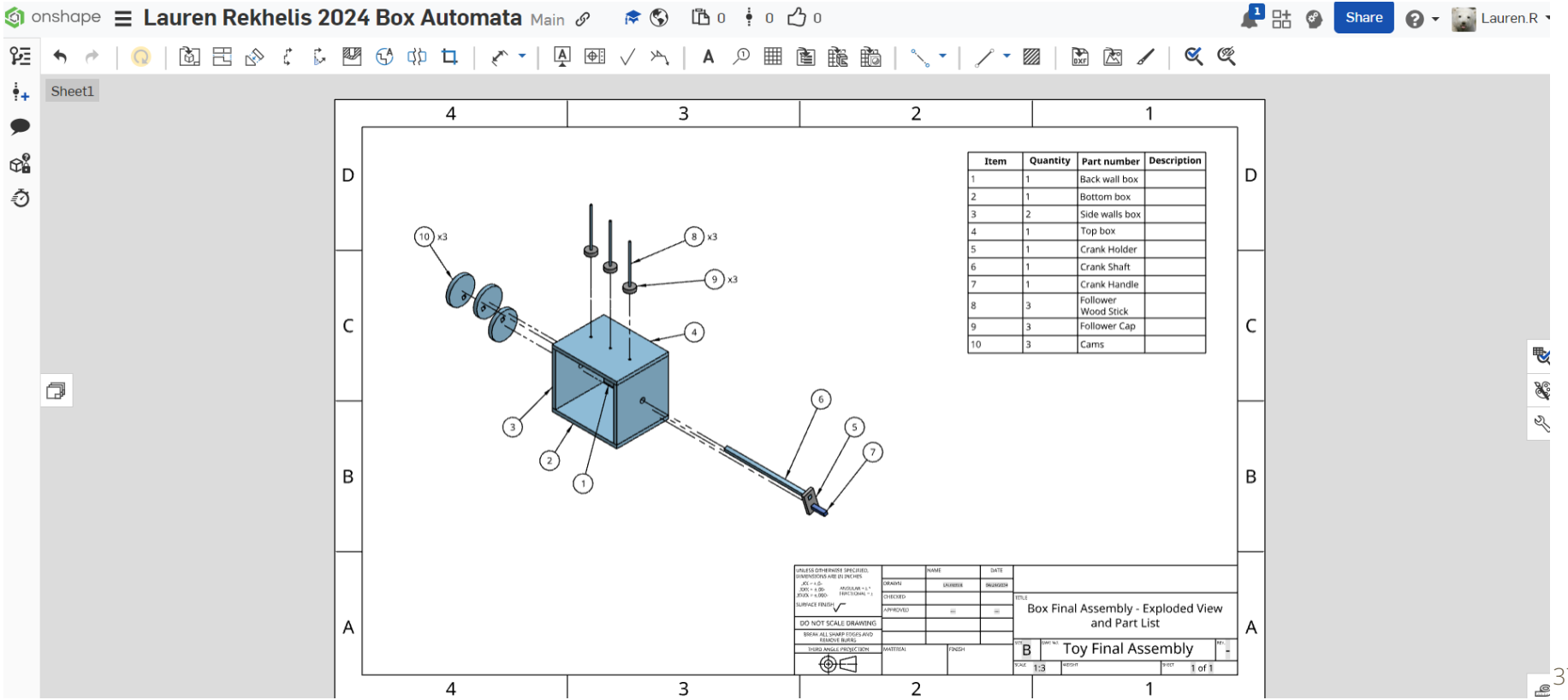
Develop a solution - Full Box Assembly

To put entire project together into one assembly

Full Box Assembly - 3D model



Full Box Assembly - Exploded view with parts BOM labeled



Construct and Test Prototype

Construct and Test Prototype

- List of materials
  - Glue used to hold all parts together (used with glue gun)
  - Wood Box (constructed from 5 rectangular pieces) to hold crank, cams, and all toys construction
  - Material to construct crank - wood square sticks
  - Knife to cut cams, stoppers and other materials.
  - Bottle plastic lids and wood sticks to construct followers
  - Plastic board to construct cams and cams holders on the crank.
  - Followers guides - used drywall plastic wall anchors
  - Pencil, cutter, other cutting tools, file tool.



Construct and Test Prototype



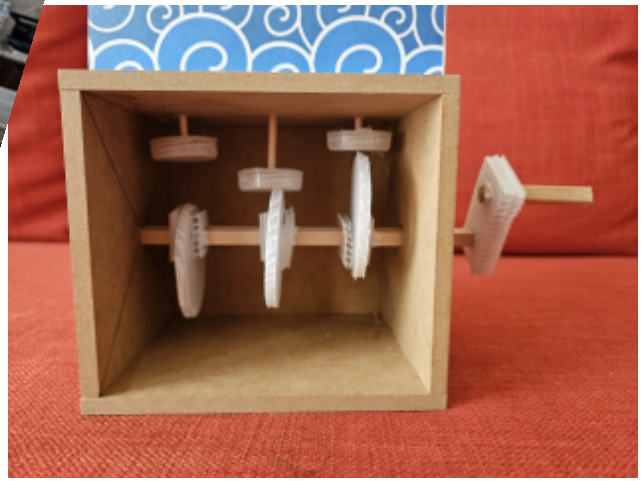




Construct and Test Prototype







Evaluate Solution

Prototype evaluation report
Review of Construction and Testing process

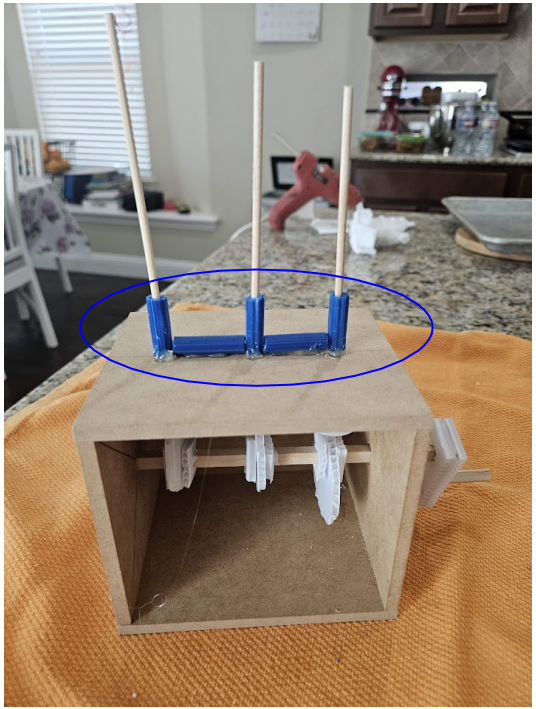
Review of Construction and Testing process

- Symmetrical construction
    - During the design I tried to create all holes in a symmetrical place from the edges to keep the toy center of gravity as much as possible close to the center. It is very important for a toy used by children knowing the fact that children balance is not always fully developed in earlier years. Keeping all holes away from edges helped during the construction to add necessary holding pieces and cups for the followers without interfering with box sides.



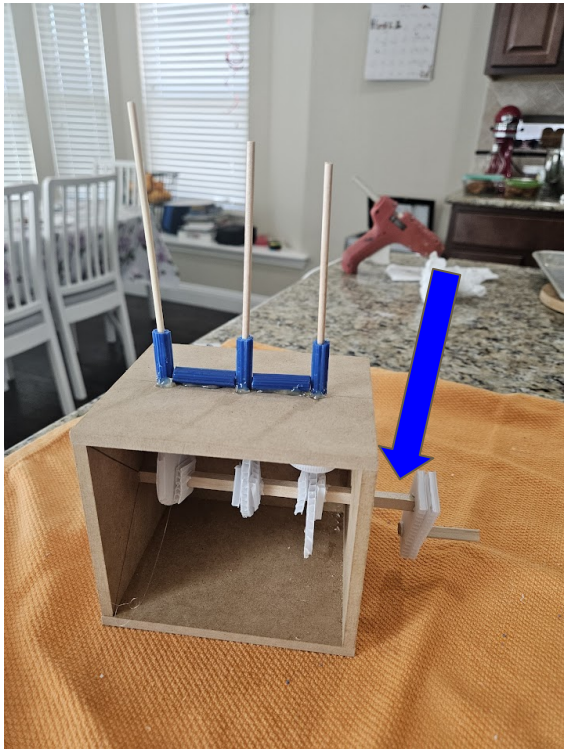
Review of Construction and Testing process (cont.)

- Challenges to keep followers balanced
  - The thickness of the box top is only 0.25 in which lead to followers not being stable. I needed to add some enhancements outside the box for stability. I constructed it from drywall plastic screw holders by creating additional tunnel for the followers. Another possible solution was to add internally to box a cylindrical tunnel for followers to be guided for longer distance than 0.25 in.
    This tunnel was attached to box. Simpler solution was selected for prototype.



Review of Construction and Testing process (cont.)

- Shaft design
  - The actual shaft that I used in design is rectangular and not cylindrical as presented in initial design. It was challenging to put them through side walls holes and make this shaft rotating smoothly. I sanded a specific part of the shaft that was rotating inside the box side walls holes in order that user (child) will be able rotate the crank smoothly. On the other side, the rectangular shaft design helped to build a more stable solution.



Review of Construction and Testing process (cont.)

- Prototyping vs. CAD design
  - Overall I was very satisfied with my CAD design which took into account required details and it led to overall good prototyping experience beside some improvements that needed to be done to make followers stable.
- Further improvements
  - For further improvement my recommendation will be to add addition cam inside in order to have additional movement device. This will allow for the designed toy to be more creative/attractive for users (children) with bigger variety of proposed designs.
    To fit additional cam will need to do scaling in cam and followers dimensions.



Present Solution

Present Solution







Thank you :-)